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Ultralow Contact Thermal Resistance between Bismuth Selenide Nanoribbons Achieved by Current-Induced Annealing

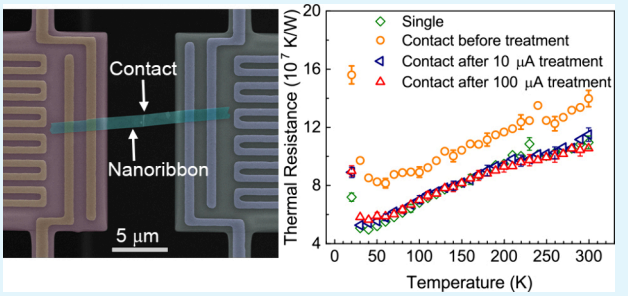
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ABSTRACT: Thermal resistance at interfaces/contacts stands as a persistent and increasingly critical issue, which hinders ultimate scaling and the performance of electronic devices. Compared to the extensive research on contact electrical resistance, contact thermal resistance and its mitigation strategies have received relatively less attention. Here, we report on an effective, in situ, and energy-efficient approach for enhancing thermal transport through the contact between semiconducting nanoribbons. By applying microampere-level electrical currents to the contact between Bi_2Se_3 nanoribbons, we demonstrate that the contact thermal resistance between two nanoribbon segments is reduced dramatically by a factor of 4, rendering the total thermal resistance of two ribbon segments with a contact approximately the same as that of the corresponding single continuous nanoribbon of the same length. Analysis suggests that the ultralow contact thermal resistance is due to enhanced phonon transmission as a result of enhanced adhesion energy at the contact, with marginal contributions from direct electron−phonon coupling, even for ohmic contacts. Our work introduces a broadly applicable electrical treatment approach to various contacts between conducting and semiconducting materials, which has important implications for the design and operation of nanoelectronic devices and energy converters.



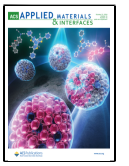
KEYWORDS: *interfacial thermal resistance, electron−phonon coupling, nanoscale contact, electrical transport, thermal transport*

1. INTRODUCTION

Thermal resistance at various contacts/interfaces represents a long-standing and increasingly critical issue for the design and operation of nanoelectronic devices.[1−4] Dated back to 1947, the invention of the point-contact junction transistor consisting of two closely placed gold foils in contact with germanium,[5] initiated the study of charge transport and the associated thermal transport through metal−semiconductor contacts. With the continuous miniaturization of electronic circuits, transistors with narrower channel widths and multi-level interconnections of finer metal lines have been under development for higher device density and better performance. Today's industrially achieved 3 nm node chips contain billions of transistors with enormous contacts all integrated into a die size of tens of square millimeters.[6] The contacts in these devices not only pose significant barriers to charge transport but also substantially impede heat dissipation.[7] Therefore, effective strategies to minimize the thermal resistance of nanoscale contacts are crucial to avoid overheat and thus ensure reliable operation of nanoelectronic devices.[2]

The numerous materials combinations and contact morphologies renders contact thermal resistance a particularly challenging issue to address. So far, various strategies to enhance thermal transport at contacts/interfaces have been developed, including the application of pressure,[8,9] chemical functionalization,[10] interlayer insertion,[11,12] and nonplanar patterning.[13,14] For instance, Losego et al. experimentally demonstrated that transitioning from van der Waals (vdW) interaction to covalent bonding *via* self-assembled monolayers could increase interfacial thermal conductance by 80% for Au/quartz interfaces.[10] Similarly, a proper choice of interlay materials has shown promising potential to enhance interfacial thermal transport.[11,12] Furthermore, increasing the overall contact area by patterning nonplanar features such as nanofabricated fin-like projections at metal/dielectric interfaces can substantially enhance interfacial thermal conductance.[13,14] These approaches for reducing contact thermal resistance require modifications of the interfaces/contacts and involve intricate deposition/etching processes, which might not be compatible with the required electrical transport properties of the contacts.

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In addition to the practical needs of minimizing contact thermal resistance, our understanding of the complex interfacial transport mechanism, especially when both electrons and phonons are involved, remains incomplete. The thermal transport at contacts/interfaces between dielectric materials is dominated by phonons, often modeled by either diffuse mismatch model (DMM) or acoustic mismatch model (AMM).[15] The situation becomes more complex at metal-nonmetal interfaces as electron−phonon interactions near/at the interface are involved.[16−18] On one hand, electrons within the metal can scatter inelastically at the interface and directly exchange energy with phonons in the dielectric, thereby establishing a direct pathway for electron−phonon coupling across the interface.[16,19] An interesting question associated with direct electron−phonon coupling at metal−semiconductor contacts/interfaces is whether the energy barrier for electrons at the contacts/interfaces substantially affects interfacial thermal conductance. Some studies suggested that lowering the barrier height could promote direct electron−phonon coupling and enhance interfacial thermal conductance,[20] while others reported no discernible effect of barrier modification on the thermal conductance.[21,22] On the other hand, electrons can also contribute to interfacial thermal transport in an indirect pathway, wherein electrons interact with phonons within metal, followed by heat conduction through the interface by phonons. Majumdar and Reddy theoretically derived a two-temperature model to quantify the contribution of this indirect electron−phonon coupling to the total thermal conductance of metal-nonmental interfaces.[17] Extensive efforts have been devoted to distinguishing the roles of direct and indirect electron−phonon coupling in interfacial thermal transport, with debates on their relative contributions. For example, the observation of comparable thermal conductance yet vastly different electronic density of states (DOS) near the Fermi energy at interfaces of Bi and Pb with diamond suggests a minimal contribution from direct electron−phonon coupling;[23] whereas theoretical analyses have argued that direct electron−phonon coupling is a viable channel for heat conduction across metal/nonmetal interfaces.[16,24] Convincing experimental evidence for the contribution of direct electron−phonon coupling to interfacial thermal transport remains elusive.

In this work, we investigated electrical and thermal transport through the contact between two bismuth selenide (Bi_2Se_3) nanoribbon segments. Bi_2Se_3 is a well-known thermoelectric material around room temperature and has been extensively adopted as nanofillers in thermoelectric nanocomposites.[25,26] The contacts/interfaces between Bi_2Se_3 nanofillers or between nanofillers and the polymer matrix can effectively suppress thermal transport. However, scattering of electrons at these contacts/interfaces could also significantly degrade electrical properties, often resulting in a lower ZT value compared to the bulk counterpart.[25] In this context, understanding electron and phonon transport through the contact between Bi_2Se_3 nanoribbons could provide insights into the transport at the contacts/interfaces in thermoelectric nanocomposites. Moreover, since both electrons and phonons contribute to thermal conductivity in Bi_2Se_3,[27] this study also provides valuable data to understand the effects of electron−phonon coupling in mediating the contact thermal resistance. Through measuring the electrical and thermal resistance of a single continuous nanoribbon and the corresponding nanoribbon segments with a contact, we experimentally show concurrently enhanced electrical and thermal transport through the contact by electrical current-induced annealing treatment, and the minimal contribution of direct electron−phonon coupling to interfacial thermal transport.

2. RESULTS AND DISCUSSION

2.1. Contact between Bi_2Se_3 Nanoribbons. Bi_2Se_3 nanoribbons were synthesized via an Au-catalyzed vapor−liquid−solid method.[27] The nanoribbons are of a rhombohedral crystal structure and grow along the $[11\bar{2}0]$ direction, as indicated by the transmission electron microscopy (TEM) micrographs (Figure S1). A ∼ 2 nm thick amorphous layer can be observed on the ribbon surface. We carefully aligned a single nanoribbon to bridge the two membranes of a suspended microdevice with a wrinkle in the middle, as denoted by the white arrow in Figure 1a. Prior to electrical and thermal measurements, an electrical current of 100 μA was applied through the two electrodes on each membrane of the microdevice to achieve ohmic contacts between the nanoribbon and the four Pt electrodes. After the measurement of a single ribbon, it is cut into two halves in the middle with a sharp probe. The wrinkle will expand naturally and form a contact between two nanoribbons, as shown in Figure 1b. The contact sample was measured again, and the electrical and thermal resistance of the contact were extracted by comparing those of the single continuous ribbon and the contact samples. Note that as the contact sample is formed in situ from the single ribbon, we assume that the contact thermal resistance between the nanoribbon and the membranes is the same for the single ribbon and contact samples. To characterize the interfacial morphology, high-resolution TEM (HRTEM) was performed to exam the contact by following the procedures in a previous study.[28] As shown in Figure 1c, a clear amorphous

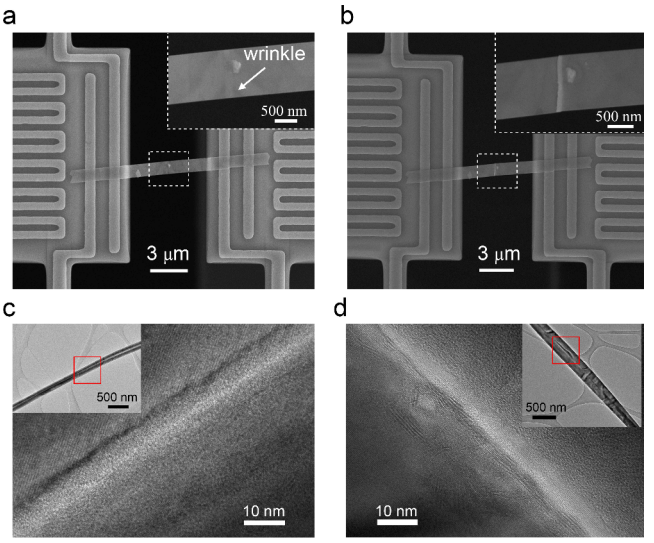


Figure 1. SEM images of (a) a single Bi_2Se_3 nanoribbon with a wrinkle and (b) the corresponding contact sample after the current-induced annealing treatment for sample S1. The inset in (b) shows the zoom-in view of the contact region. A contact between the nanoribbons is clearly observed. Representative HRTEM images of the contact (c) before and (d) after the current-induced annealing treatment. The insets show the corresponding low-magnification TEM images, where the red boxes indicate the interfacial region selected for HRTEM characterization.

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interface with a thickness of ~5 nm can be observed at the contact.

2.2. Contact Electrical Resistance. Figure 2a and 2b plot the measured electrical and thermal resistance of sample S1.

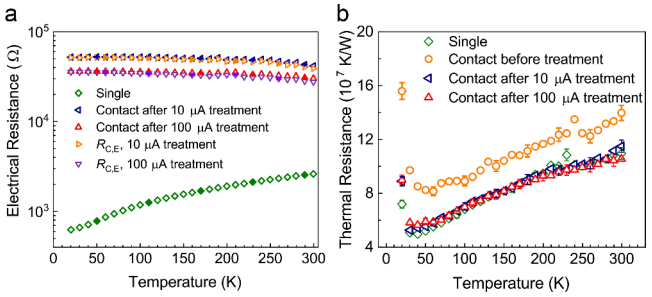


Figure 2. (a) Measured electrical and (b) thermal resistance of the single nanoribbon and the corresponding contact sample for sample S1.

The electrical resistance of the single nanoribbon exhibits a monotonic increasing trend with temperature, indicating a semimetallic behavior. The thermal resistance of the single nanoribbon decreases rapidly at low temperatures, reaches a minimum at approximately 30−40 K, followed by an increasing trend as temperature further rises. The temperature dependence trend of thermal resistance is consistent with those typically observed for single-crystalline materials.[27] The electron contribution to the thermal conductivity was estimated using the Wiedemann−Franz law, allowing for the separation of the lattice thermal conductivity. At room temperature, the electronic thermal conductivity accounts for 30% of the total thermal conductivity (Figure S2). For the contact sample without the current-induced annealing treatment, no electrical current can be detected in the current−voltage measurement when a small voltage is applied (Figure S3a). The thermal resistance of the contact sample shows a similar temperature trend to that of the single ribbon but with a higher value, suggesting that the contact introduces a substantial resistance to thermal transport.

To activate the electrical contact between the nanoribbons, a relatively large electrical current of 10 μA was passed through the contact sample. Specifically, the sample was placed in a vacuum chamber maintained at a pressure below 10^{-6} mbar. For 10 μA electrical current-induced annealing, the sample was connected in series with a 1 MΩ resistor. The applied voltage gradually ramps up from 0 to 10 V with a step increment of 1 V. The entire process was completed within 10 min. For the 100 μA electrical current treatment, the 1 MΩ resistor was replaced by a 100 kΩ resistor. Compared to conventional thermal annealing, this electrical current-induced annealing approach offers the advantages of localized annealing, particularly for high-resistance nanoscale contacts, and ease of implementation. As shown in Figure S3b, a linear current−voltage curve was observed after this current-induced annealing treatment, indicating the formation of an ohmic contact between the nanoribbons. The measured electrical resistance of the contact sample is over 1 order of magnitude higher than that of the single nanoribbon sample and only varies slightly with temperature (Figure 2a), indicating that the electrical resistance is dominated by the contact. The contact electrical resistance between two nanoribbons is extracted by comparing the electrical resistances of the contact sample and the single ribbon, which nearly coincides with the electrical resistance of the contact sample (Figure 2a). Interestingly, the thermal resistance of the contact sample after the electrical treatment is reduced and nearly overlaps with that of the single nanoribbon sample (Figure 2b). To further investigate the effects of the annealing electrical current on electrical and thermal transport through the contact, a higher electrical current of 100 μA was applied to the contact sample. This treatment further reduces the electrical resistance of the contact sample, while the thermal resistance remains nearly unchanged (Figure 2b). Notably, after the current-induced annealing treatment, a clear contact edge can still be observed in the scanning electron microscopy (SEM) micrograph as shown in the inset of Figure 1b. To examine the effect of the current-induced annealing treatment on the interfacial morphology, HRTEM was conducted at the contact after electrical treatment. As shown in Figure 1d, no obvious modification in the amorphous interface can be identified.

The temperature-independent contact electrical resistance (Figure 2a) suggests that electron transport through the contact is likely dominated by field emission.[29,30] According to the field emission model,[29] the specific contact electrical resistance can be calculated by

$$R_C = \left[\frac{A^*T\pi q}{k_B \sin(\pi c k_B T)} \exp\left(\frac{-q\phi_b}{E_{00}}\right) - \frac{A^* q}{c k_B^2} \exp\left(\frac{-q\phi_b}{E_{00}} - cE_F\right)\right]^{-1} \propto \exp\left(\frac{q\phi_b}{E_{00}}\right) \quad (1)$$

where $A^* = \frac{4\pi q m^* k_B^2}{h^3}$ is the Richardson constant, T is the absolute temperature, q is the electronic charge, and k_B is the Boltzmann's constant. $c = \frac{1}{2E_{00}} \ln\left(\frac{4q\phi_b}{E_F}\right)$, where ϕ_b is the barrier height and E_F is the Fermi energy with respect to the energy band edge in the semiconductor bulk. $E_{00} = \frac{qh}{4\pi}\sqrt{\frac{n}{\varepsilon_S m^*}}$ represents a characteristic energy related to the tunneling probability, which depends on the doping concentration n, the semiconductor permittivity ε_S, and the effective mass of tunneling electron m^*. For the contact sample before the current-induced annealing treatment, under a small applied voltage, the electron energy is too low to overcome the energy barrier at the contact. The current-induced annealing can effectively reduce the barrier height,[31] leading to an ohmic contact with reduced contact electrical resistance (Figure 2a). The temperature-independent contact electrical resistance results from the fact that the energy barrier height and the carrier concentration of Bi_2Se_3 are not sensitive to temperature.[32]

2.3. Contact Thermal Resistance. Interestingly, while electrical transport in the annealed sample is remarkably suppressed by the contact compared to the single ribbon, the measured total thermal resistance of the contact sample is almost the same as that of the single ribbon sample (Figure 2b). This observation indicates that the current-induced annealing effectively minimizes the contact thermal resistance to a negligible level. Importantly, measurements on two additional samples yielded consistent results (Figure S4 and S5).

In Bi_2Se_3 nanoribbons, both electrons and phonons contribute to heat conduction, with the electron contribution accounting for 30% of the total thermal conductivity at room

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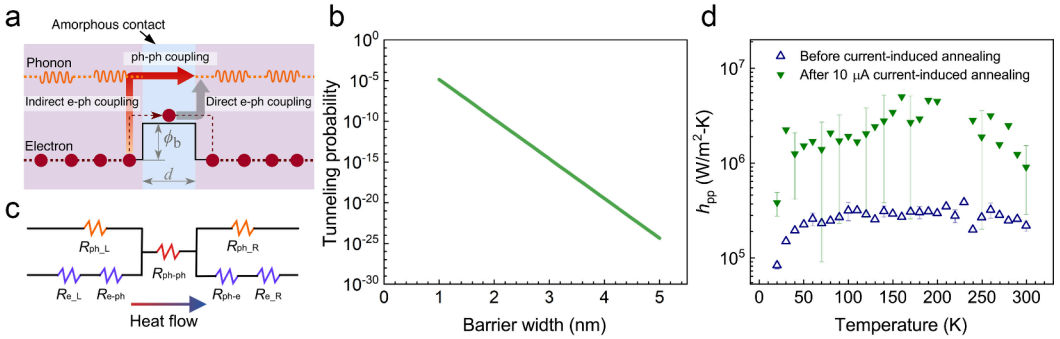


Figure 3. (a) Schematic of electron and phonon transport through the contact between Bi_2Se_3 nanoribbons. Three heat conduction mechanisms are included: indirect electron−phonon coupling ($R_{e\text{-}ph}$ and $R_{ph\text{-}e}$), direct electron−phonon coupling (R_{ei}), and phonon−phonon coupling ($R_{ph\text{-}ph}$). (b) Electron tunneling probability through the contact between Bi_2Se_3 nanoribbons as a function of the barrier width. (c) Equivalent thermal circuit of the contacted Bi_2Se_3 nanoribbons where direct electron−phonon coupling is neglected. R_e and R_{ph} are electronic and phononic thermal resistance of the single nanoribbon, respectively. The subscripts L and R denote the left and right ribbons, respectively. (d) Extracted phonon thermal conductance per unit area (h_{pp}) for the contact between the nanoribbons before and after the current-induced annealing treatment.

temperature. At the contact, where electron transport is significantly suppressed, there are three potential pathways for thermal transport as illustrated schematically in Figure 3a. One pathway is phonon−phonon coupling with a thermal resistance of $R_{ph\text{-}ph}$. Other pathways related to electron contributions include direct electron−phonon coupling *via* inelastic scattering with a thermal resistance of R_{ei}, and indirect electron−phonon coupling with a thermal resistance $R_{e\text{-}ph}$, followed by phonon−phonon transport across the contact. Note that the electron−electron pathway has been neglected due to the marginal value predicted for this channel based on the Wiedemann−Franz law.

Subsequently, we evaluate the relative contribution of different pathways to the overall contact thermal conductance. For direct electron−phonon coupling channel, only electrons tunneling through the barrier can exchange energy with phonons on the opposite side. Therefore, the tunneling probability serves as an indicator of the contribution from direct electron−phonon coupling at the contact. For a rectangular barrier with an effective barrier height ϕ_b and width d (corresponding to the thickness of interfacial amorphous layer), the tunneling probability can be expressed as $T_r \approx \exp\left(-\frac{4\pi d\sqrt{2qm^*\phi_b}}{h}\right)$.[33] Employing the thermoelectric properties measured for a single nanoribbon sample, first-principles calculations were applied to calculate the material parameters of Bi_2Se_3, as listed in Table 1. Combining the measured contact electrical resistance and eq 1, ϕ_b is determined to be 0.68 eV. Figure 3b plots T_r as a function of d. T_r is approximately 1.4×10^{-5} when d is merely 1 nm and it drops exponentially to 4.5×10^{-25} as d increases to 5 nm. The HRTEM images in Figure 1c and 1d show that the total thickness of the interfacial amorphous layer is approximately 5 nm, which corresponds to a negligible tunneling probability and thus minimal contribution of direct electron−phonon coupling to interfacial thermal transport from electron tunneling.

Table 1. Material Properties of Bi_2Se_3[a]

N (m^{-3})	E_F (eV)	m^*	ε_S	ρ (kg/m^3)
7.27×10^{24}	0.106	$1.77m_0$	$23.2\varepsilon_0$	7681

[a] m_0 and ε_0 are the free-electron mass and the dielectric constant of free space, respectively.

Moreover, recent first-principles calculations have suggested that direct electron−phonon coupling between electrons in the metal and phonons in the semiconductor could be mediated by delocalized phonon modes whose vibrational energy is distributed across the interface. In contrast, localized phonon modes in the semiconductor region of the interface exhibit negligible coupling with electrons in the metal, indicating that the direct electron−phonon coupling is intimately tied to the strength of interfacial bonding.[34] Importantly, the length scale of electron−phonon interaction (the spatial extent of joint interface modes) is very sensitive to the interfacial bonding strength.[34] Typically, this length ranges from approximately 1−2 nm for interfaces with covalent bonding to nearly zero for vdW interfaces.[34−36] For the vdW contact between Bi_2Se_3 nanoribbons with a 5 nm thick interfacial amorphous layer, the contribution from delocalized phonon modes is expected to be minimal. Instead, the phonon modes will be localized on either side of the contact, which cannot sustain effective direct electron−phonon coupling. Taken together, we believe that direct electron−phonon coupling does not make any significant contribution to thermal transport through the contact between Bi_2Se_3 nanoribbons.

Now considering indirect electron−phonon coupling, the overall thermal resistance of the contact sample can be expressed as

$$R_{tot} = [(R_{e_L} + R_{e-ph})//R_{ph_L}] + R_{ph-ph} + [(R_{ph-e} + R_{e_R})//R_{ph_R}]$$

as illustrated in Figure 3c, where R_e and R_{ph} represent electron and phonon thermal resistance of a single nanoribbon segment, respectively. The subscripts L and R indicate left and right ribbon segments, respectively. The symbol "//" denotes a parallel connection of resistors. The thermal resistance of the overlapped region is approximated as the sum of $R_{ph\text{-}ph}$ and the thermal resistance of a single ribbon segment with one-half of the overlapped length. According to the two-temperature model,[17] the effective interfacial thermal conductance of indirect electron−phonon coupling (the reciprocal of the serial resistances $R_{e\text{-}ph}$ and $R_{ph\text{-}ph}$) is calculated as[17]

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$$h^* = \frac{\kappa_{ph}\sqrt{\frac{G}{\kappa_m}}}{\left[1 + \frac{\kappa_{ph}}{h_{pp}}\sqrt{\frac{G}{\kappa_m}} - \frac{\kappa_m}{\kappa_e}\right]}$$

where $\kappa_m = \left(\frac{1}{\kappa_e} + \frac{1}{\kappa_{ph}}\right)^{-1}$ is the harmonic mean of the lattice thermal conductivity (κ_{ph}) and the electronic thermal conductivity (κ_e), which are measured to be 1.06 and 0.47 W/m·K, respectively, at room temperature (Figure S2). h_{pp} is the phonon−phonon interfacial thermal conductance per unit area. G is the electron−phonon coupling constant, which is assumed to be independent of temperature with a value of 1.53 × 10^{16} W/m^3·K.[37] By combining the measured thermal resistance of the single ribbon and contact samples, h_{pp} before and after the current-induced annealing treatment was extracted and plotted in Figure 3d. The relative uncertainty in h_{pp} at room temperature is estimated to be 22% and 68% for the contact before and after the current-induced annealing treatment, respectively (Section V in the Supporting Information). h_{pp} first increases and then levels off with temperature. Before the current-induced annealing treatment, h_{pp} is approximately 2.2 × 10^5 W/m^2·K at 300 K, which is comparable to the corresponding value for the contact between copper phthalocyanine nanoribbons[38] but approximately 2 orders of magnitude lower than those of the transfer-printed interfaces.[39,40] The ultralow h_{pp} may be attributed to the rather loose contact morphology through weak vdW interactions between Bi_2Se_3 nanoribbons.[38] After current-induced annealing treatment, h_{pp} is significantly enhanced, reaching ∼9.1 × 10^5 W/m^2·K at 300 K, which is about 4-fold of the value before annealing treatment. It should be noted that the enhanced contact thermal conductance is still much lower than that of high-quality interfaces with strong bonding strength,[23,41] primarily due to the limited adhesion energy for vdW contacts between the Bi_2Se_3 nanoribbons, even after current-induced annealing treatment. Importantly, it has been shown that for vdW contacts between boron nanoribbons, the adhesion energy could be >300 mJ/m^2, which should correspond to much high contact thermal conductance.[42] The enhanced contact thermal conductance is reasonable as current-induced annealing could improve the adhesion between the two ribbon segments through local heating as electrons are forced through the high resistance contact. The enhanced adhesion facilitates phonon transmission and thermal transport through the contact.

The contact thermal resistance between Bi_2Se_3 nanoribbons consists of the thermal resistance due to the vdW contact, which depends on the adhesion energy, and the thermal resistance of the amorphous layer. The thermal resistance due to the vdW contact (R_{vdw}) can be estimated by

$$R_{vdW} = \left[\frac{1}{2}\sum_j \int_0^{\pi/2}\int_0^{\omega_{m,j}} d\omega \frac{dN_j(\omega, T)}{dT}\hbar\omega v_j \tau_{\omega,j} \sin\theta \cos\theta \, d\theta \, d\omega\right]^{-1} \quad (2)$$

where j represents the phonon branch, ω is the phonon angular frequency, ω_m is the maximum phonon angular frequency related to the Debye temperature θ_D (136.9 K based on first-principles calculations) by $\omega_{m,j} = \frac{k_B\theta_{D,j}}{\hbar}$, v is phonon group velocity (2900 m/s for longitudinal and 1700 m/s for transverse phonons[43]), $N_j = \frac{\omega^2}{2\pi^2 v_j^2[\exp(\hbar\omega / k_B T) - 1]}$ is phonon distribution function, θ is the angle between the interface normal and the phonon propagation direction in the crystalline Bi_2Se_3, and $\tau_{\omega,j}$ is the phonon transmission coefficient through the vdW contact that can be written as[15,44,45]

$$\tau_{\omega,j} = \frac{1}{1 + \left(\frac{\omega_j^2}{2K_A^2}\right)z_j^2 \cos^2\theta} \quad (3)$$

Here, z is the acoustic impedance (equal to the product of density and phonon group velocity, ρv) and K_A is the spring constant per unit area related to the interfacial adhesion energy E_{ad} (Section VII in the Supporting Information). The thermal resistance of the amorphous layer can be simply estimated by $R_{am} = d/\kappa$, where d is the thickness of the amorphous layer taken as 5 nm, and κ is its thermal conductivity. At room temperature, κ is expected to lie between the amorphous limit (0.27 W/m-K)[46] and the phononic thermal conductivity of the Bi_2Se_3 nanoribbons (1.07 W/m-K). The contact thermal conductance between nanoribbons is given by $h_{pp} = (R_{vdW} + R_{am})^{-1}$. Note that the estimated R_{am} is in the range of 0.47−1.85 × 10^{-8} m^2·K/W, which only contribute ∼1% to the interfacial thermal resistance. Neglecting R_{am}, the only unknown parameter in calculating h_{pp} is E_{ad}. To illustrate the effect of E_{ad} on h_{pp}, the room-temperature h_{pp} was calculated as a function of E_{ad} and plotted in Figure 4a. As expected, h_{pp} increases monotonically with E_{ad}. Based on measured h_{pp}, E_{ad} was determined to be 23 mJ/m^2 and 52 mJ/m^2 for the contact before and after current-induced annealing, respectively. The corresponding R_{vdW} is 4.46 × 10^{-6} m^2·K/W and 1.09 × 10^{-6} m^2·K/W, respectively, before and after the current-induced annealing, which is indeed much higher than the thermal resistance of the amorphous layer. This understanding is also consistent with our experimental results that current-induced annealing effectively reduces the contact thermal resistance between Bi_2Se_3 nanoribbons by enhancing the adhesion energy, which cannot be the case if the thermal resistance of the amorphous layer dominates the contact thermal resistance. For vdW contacts between semiconducting/conducting nanostructures, where an electrical current can be applied, current-induced annealing could be an effective method to enhance the adhesion energy and consequently the contact thermal conductance. The adhesion energy between two ideal Bi_2Se_3 nanoribbons with a thin amorphous layer was also

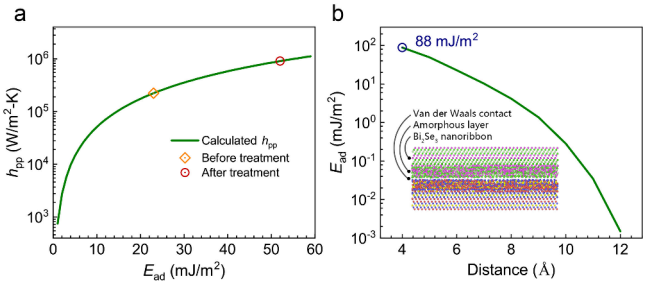


Figure 4. (a) Adhesion energy derived from the measured interfacial thermal conductance before and after current-induced annealing. (b) Calculated adhesion energy as a function of the separation distance. The inset shows the model system used in MD simulations. Two ideal Bi_2Se_3 nanoribbons with a thin amorphous layer form a van der Waals contact.

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calculated using molecular dynamics (MD) simulations (Section VIII in the Supporting Information). As shown in Figure 4b, the adhesion energy monotonically decreases with the separation distance between two nanoribbons. At the equilibrium separation distance of approximately 4 Å, the adhesion energy is calculated to be 88 mJ/m^2, which is slightly larger the value derived for the contact after electrical current-induced annealing (52 mJ/m^2). It is worth noting that the model system in MD simulations represents an ideal contact between two Bi_2Se_3 nanoribbons, whereas experimentally measured nanoribbons inevitably have surface imperfections that could reduce the adhesion energy.

Figure 5a presents the measured Seebeck coefficient of the single continuous ribbon and the contact sample, respectively.

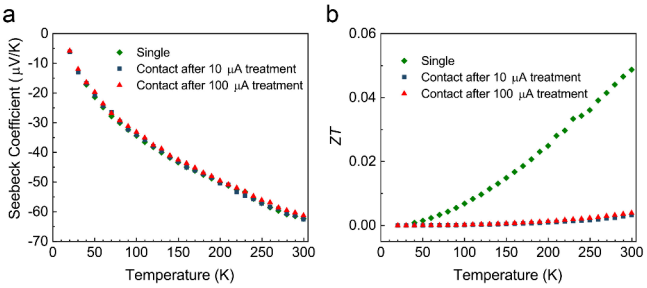


Figure 5. (a) Measured Seebeck coefficient and (b) ZT of the single nanoribbon and the contact sample after the treatment with an electrical current of 10 and 100 μA for sample S1.

The overlapping curves indicate that the Seebeck coefficient is not affected by the contact. The effective Seebeck coefficient of the contact sample can be expressed as $S^* = S_{NR} + \gamma(S_C - S_{NR})$, where S_{NR}, S_C, and γ are the Seebeck coefficient of the nanoribbon, the Seebeck coefficient of the contact, and the ratio of the temperature drop across the contact to the total temperature drop between heat source and heat sink, respectively.[47] Given that the thermal resistance of the contact sample is nearly the same as that of the single continuous ribbon (Figure 2b), the temperature drop across the contact is marginal compared to the total temperature drop. Consequently, γ is approximately zero, and S^* is equal to S_{NR}. Figure 5b shows that the ZT of the contact sample is dramatically lower than that of the single nanoribbon sample, primarily due to the substantial contact electrical resistance between the nanoribbons. These results on the single nanoribbon and the contact sample could help explain the suppressed thermoelectric properties of nanowire/nanoribbon composites compared to their constituent counterparts.[25,26]

It is instructive to compare electrical and thermal transport through the contacts between semiconductor nanostructures with transport phenomena through the interfaces/boundaries in nanostructured thermoelectric materials. Interfaces/boundaries in nanostructured thermoelectric materials, usually formed by direct growth,[48] sintering,[49] or hot pressing,[50,51] impede thermal transport more severely than electrical transport, thereby enhancing thermoelectric properties. Meanwhile, the gain size in these materials is comparable to the phonon MFP but much larger than the electron MFP, leading to more intense phonon scattering than electron scattering. In contrast, for the contact between semiconductor nanostructures, the weak vdW interaction and the presence of an interfacial amorphous layer impose a substantial energy barrier for electron transport. Despite partial reduction of the barrier height through current-induced annealing, only a marginal portion of electrons can tunnel through the contact, resulting in significant contact electrical resistance. Those electrons blocked by the contact fail to contribute to electrical transport but engage in energy exchange with phonons. As current-induced annealing effectively strengthens the adhesion between the nanoribbon segments, the phonon−phonon conduction path is enhanced, yielding a total thermal resistance comparable to that of the single ribbon counterpart with the assistance of indirect electron−phonon coupling.

3. CONCLUSION

In summary, we report on a substantial 4-fold reduction in the contact thermal resistance between Bi_2Se_3 nanoribbons achieved by current-induced annealing, which is attributed to enhanced adhesion energy at the contact. In contrast, electrical transport in the contact sample is largely hindered by the considerable resistance associated with the energy barrier posed by the contact. Moreover, we show that direct electron−phonon coupling plays a marginal role in thermal transport even through ohmic contact is formed. This work provides a facile approach for reducing thermal resistance at contacts between conducting/semiconducting materials, which offers insights into thermal management for nanoelectronic devices and the design of thermoelectric converters.

4. METHODS

4.1. Materials Synthesis and Characterization. In this work, Bi_2Se_3 nanoribbons were synthesized *via* an Au-catalyzed vapor−liquid−solid method as described previously.[27] The crystalline structure of the synthesized nanoribbons was characterized by HRTEM as shown in Figure S1, while the chemical composition was examined by HRTEM/energy dispersive X-ray spectroscopy. TEM characterization confirms that Bi_2Se_3 nanoribbons have a rhombohedral crystal structure and a growth direction of $[1\bar{1}20]$.

4.2. Thermal and Electrical Measurements. Thermal and electrical resistance of the samples were measured over a temperature range from 20 to 300 K by using a microthermal bridge method and a four-probe method, respectively.[52,53] As shown in Figure 1, the measurement microdevice consists of two suspended membranes. Serpentine Pt coils were patterned on these membranes, serving as heaters/resistance thermometers for thermal measurement. Two separate Pt electrodes were also fabricated on each membrane, which enables the four-probe electrical measurement. The electrical resistance was measured by passing an electrical current of less than 1 μA through the sample to ensure that the Joule heating effect is minimal. To eliminate the convective heat loss, all measurements were performed in a cryostat system (Janis CCS-400H/204) with a high vacuum (<10^{-6} mbar). The suspended length and width of the measured Bi_2Se_3 nanoribbons were characterized by SEM, and the thickness was measured by atomic force microscopy.

4.3. First-Principles Calculations. All first-principles calculations are conducted using the Quantum ESPRESSO package. We adopt the PBEsol-GGA exchange-correlation function and optimized norm-conserving Vanderbilt (ONCV) pseudopotentials provided by the PseudoDojo project.[54] The lattice parameters are obtained from the experiment, namely, a = 4.138 Å and c = 28.64 Å, which can generate more reasonable phonon dispersion compared to the lattice constants optimized by first-principles calculation.[55] The atomic positions are fully relaxed based on the Broyden-Fretcher-Goldfarb-Shanno (BFGS) optimization method.[56−58] The energy cutoff of the plane wave is set to be 60 Ry and the convergence threshold of electron energy is 10^{-10} Ry in the self-consistent field calculation. The **k**-point grid is set to be 6 × 6 × 6, and the **q**-point grid is set to be 4 × 4 × 4 in the density functional perturbation theory calculation. The phonon

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dispersion is shown in Figure S6. The Debye temperature is calculated to be 136.9 K using the phonon density of states.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.4c10789.

TEM and HRTEM images of a Bi_2Se_3 nanoribbon, electronic and phononic thermal conductivity of the Bi_2Se_3 nanoribbon, current−voltage measurements of the single nanoribbon and the contact sample before and after the current-induced annealing treatments with different currents, additional experimental results, uncertainty analysis, phonon dispersion of Bi_2Se_3, calculation of the phonon transmission coefficient, calculation of the adhesion energy (PDF)

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Author Contributions

Y.X. conducted transport measurements and theoretical modeling. S.L. performed first-principles calculations. R.W. performed the HRTEM characterization. Y.X. prepared the manuscript. D.L. and D.X. edited the manuscript. All authors discussed the results and commented on the manuscript.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Moore, A. L.; Shi, L. Emerging challenges and materials for thermal management of electronics. *Mater. Today* **2014**, *17*, 163−174.

(2) Warzoha, R. J.; Wilson, A. A.; Donovan, B. F.; Donmezer, N.; Giri, A.; Hopkins, P. E.; Choi, S.; Pahinkar, D.; Shi, J.; Graham, S.; et al. Applications and Impacts of Nanoscale Thermal Transport in Electronics Packaging. *J. Electron. Packag.* **2021**, *143*, 020804.

(3) Lee, C.-H.; Lee, G.-H.; van der Zande, A. M.; Chen, W.; Li, Y.; Han, M.; Cui, X.; Arefe, G.; Nuckolls, C.; Heinz, T. F.; Guo, J.; Hone, J.; Kim, P. Atomically thin p−n junctions with van der Waals heterointerfaces. *Nat. Nanotechnol.* **2014**, *9*, 676−681.

(4) Roy, A. K.; Farmer, B. L.; Varshney, V.; Sihn, S.; Lee, J.; Ganguli, S. Importance of Interfaces in Governing Thermal Transport in Composite Materials: Modeling and Experimental Perspectives. *ACS Appl. Mater. Interfaces* **2012**, *4*, 545−563.

(5) Bardeen, J.; Brattain, W. H. The Transistor, A Semi-Conductor Triode. *Phys. Rev.* **1948**, *74*, 230−231.

(6) Kalinin, S. V.; Ostertak, D. I.; Poteryaev, D. A.; Kuznetsov, M. A. A New Revolution in Logic Silicon IC Technology: GAA FETs are Replacing FinFETs. *2023 IEEE XVI Int. Sci. Tech. Conf. Actual Probl. Electron. Instrum. Eng.* **2023**, 160−165, DOI: 10.1109/APEIE59731.2023.10347677.

(7) Pop, E.; Goodson, K. E. Thermal Phenomena in Nanoscale Transistors. *J. Electron. Packag.* **2006**, *128*, 102−108.

(8) Hohensee, G. T.; Wilson, R. B.; Cahill, D. G. Thermal conductance of metal-diamond interfaces at high pressure. *Nat. Commun.* **2015**, *6*, 6578.

(9) Hsieh, W.-P.; Lyons, A. S.; Pop, E.; Keblinski, P.; Cahill, D. G. Pressure tuning of the thermal conductance of weak interfaces. *Phys. Rev. B* **2011**, *84*, 184107.

(10) Losego, M. D.; Grady, M. E.; Sottos, N. R.; Cahill, D. G.; Braun, P. V. Effects of chemical bonding on heat transport across interfaces. *Nat. Mater.* **2012**, *11*, 502−506.

(11) Li, X.; Park, W.; Wang, Y.; Chen, Y. P.; Ruan, X. Reducing interfacial thermal resistance between metal and dielectric materials by a metal interlayer. *J. Appl. Phys.* **2019**, *125*, 045302.

(12) Chang, G.; Sun, F.; Wang, L.; Che, Z.; Wang, X.; Wang, J.; Kim, M. J.; Zhang, H. Regulated Interfacial Thermal Conductance between Cu and Diamond by a TiC Interlayer for Thermal Management Applications. *ACS Appl. Mater. Interfaces* **2019**, *11*, 26507−26517.

(13) Lee, E.; Zhang, T.; Yoo, T.; Guo, Z.; Luo, T. Nanostructures Significantly Enhance Thermal Transport across Solid Interfaces. *ACS Appl. Mater. Interfaces* **2016**, *8*, 35505−35512.

(14) Park, W.; Sood, A.; Park, J.; Asheghi, M.; Sinclair, R.; Goodson, K. E. Enhanced Thermal Conduction Through Nanostructured Interfaces. *Nanoscale Microscale Thermophys. Eng.* **2017**, *21*, 134−144.

(15) Swartz, E. T.; Pohl, R. O. Thermal boundary resistance. *Rev. Mod. Phys.* **1989**, *61*, 605−668.

(16) Sergeev, A. V. Electronic Kapitza conductance due to inelastic electron-boundary scattering. *Phys. Rev. B* **1998**, *58*, No. R10199.

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(17) Majumdar, A.; Reddy, P. Role of electron-phonon coupling in thermal conductance of metal-nonmetal interfaces. *Appl. Phys. Lett.* **2004**, *84*, 4768−4770.
(18) Li, Z.; Tan, S.; Bozorg-Grayeli, E.; Kodama, T.; Asheghi, M.; Delgado, G.; Panzer, M.; Pokrovsky, A.; Wack, D.; Goodson, K. E. Phonon Dominated Heat Conduction Normal to Mo/Si Multilayers with Period below 10 nm. *Nano Lett.* **2012**, *12*, 3121−3126.
(19) Huberman, M. L.; Overhauser, A. W. Electronic Kapitza conductance at a diamond-Pb interface. *Phys. Rev. B* **1994**, *50*, 2865−2873.
(20) Horny, N.; Chirtoc, M.; Fleming, A.; Hamaoui, G.; Ban, H. Kapitza thermal resistance studied by high-frequency photothermal radiometry. *Appl. Phys. Lett.* **2016**, *109*, 033103.
(21) Ye, N.; Feser, J. P.; Sadasivam, S.; Fisher, T. S.; Wang, T.; Ni, C.; Janotti, A. Thermal transport across metal silicide-silicon interfaces: An experimental comparison between epitaxial and nonepitaxial interfaces. *Phys. Rev. B* **2017**, *95*, 085430.
(22) Hamaoui, G.; Horny, N.; Hua, Z.; Zhu, T.; Robillard, J.-F.; Fleming, A.; Ban, H.; Chirtoc, M. Electronic contribution in heat transfer at metal-semiconductor and metal silicide-semiconductor interfaces. *Sci. Rep.* **2018**, *8*, 11352.
(23) Lyeo, H.-K.; Cahill, D. G. Thermal conductance of interfaces between highly dissimilar materials. *Phys. Rev. B* **2006**, *73*, 144301.
(24) Sadasivam, S.; Waghmare, U. V.; Fisher, T. S. Electron-phonon coupling and thermal conductance at a metal-semiconductor interface: First-principles analysis. *J. Appl. Phys.* **2015**, *117*, 134502.
(25) Dun, C.; Hewitt, C. A.; Huang, H.; Xu, J.; Montgomery, D. S.; Nie, W.; Jiang, Q.; Carroll, D. L. Layered Bi_2Se_3 Nanoplate/Polyvinylidene Fluoride Composite Based n-type Thermoelectric Fabrics. *ACS Appl. Mater. Interfaces* **2015**, *7*, 7054−7059.
(26) Dun, C.; Hewitt, C. A.; Huang, H.; Xu, J.; Zhou, C.; Huang, W.; Cui, Y.; Zhou, W.; Jiang, Q.; Carroll, D. L. Flexible n-type thermoelectric films based on Cu-doped Bi_2Se_3 nanoplate and Polyvinylidene Fluoride composite with decoupled Seebeck coefficient and electrical conductivity. *Nano Energy* **2015**, *18*, 306−314.
(27) Tang, H.; Wang, X. M.; Xiong, Y. C.; Zhao, Y.; Zhang, Y.; Zhang, Y.; Yang, J. K.; Xu, D. Y. Thermoelectric characterization of individual bismuth selenide topological insulator nanoribbons. *Nanoscale* **2015**, *7*, 6683−6690.
(28) Yang, L.; Zhang, Q.; Cui, Z.; Gerboth, M.; Zhao, Y.; Xu, T. T.; Walker, D. G.; Li, D. Ballistic Phonon Penetration Depth in Amorphous Silicon Dioxide. *Nano Lett.* **2017**, *17*, 7218−7225.
(29) Yu, A. Y. C. Electron tunneling and contact resistance of metal-silicon contact barriers. *Solid-State Electron.* **1970**, *13*, 239−247.
(30) Schroder, D. K.; Meier, D. L. Solar cell contact resistance—A review. *IEEE Trans. Electron Devices* **1984**, *31*, 637−647.
(31) Iucolano, F.; Roccaforte, F.; Alberti, A.; Bongiorno, C.; Di Franco, S.; Raineri, V. Temperature dependence of the specific resistance in Ti/Al/Ni/Au contacts on n-type GaN. *J. Appl. Phys.* **2006**, *100*, 123706.
(32) Amaladass, E. P.; Devidas, T. R.; Sharma, S.; Sundar, C. S.; Mani, A.; Bharathi, A. Magneto-transport behaviour of $Bi_2Se_{3-x}Te_x$: role of disorder. *J. Phys.: Condens. Matter* **2016**, *28*, 075003.
(33) Sze, S. M.; Ng, K. K. *Physics of Semiconductor Devices*; Wiley: New York, 2006.
(34) Sadasivam, S.; Ye, N.; Feser, J. P.; Charles, J.; Miao, K.; Kubis, T.; Fisher, T. S. Thermal transport across metal silicide-silicon interfaces: First-principles calculations and Green's function transport simulations. *Phys. Rev. B* **2017**, *95*, 085310.
(35) Lin, K.-H.; Strachan, A. Role of direct electron-phonon coupling across metal-semiconductor interfaces in thermal transport *via* molecular dynamics. *J. Chem. Phys.* **2015**, *143*, 034703.
(36) Lu, Z.; Wang, Y.; Ruan, X. Metal/dielectric thermal interfacial transport considering cross-interface electron-phonon coupling: Theory, two-temperature molecular dynamics, and thermal circuit. *Phys. Rev. B* **2016**, *93*, 064302.
(37) Lai, Y.-P.; Chen, H.-J.; Wu, K.-H.; Liu, J.-M. Temperature-dependent carrier-phonon coupling in topological insulator Bi_2Se_3. *Appl. Phys. Lett.* **2014**, *105*, 232110.
(38) Xiong, Y.; Yu, X.; Huang, Y.; Yang, J.; Li, L.; Yang, N.; Xu, D. Ultralow thermal conductance of the van der Waals interface between organic nanoribbons. *Mater. Today Phys.* **2019**, *11*, 100139.
(39) Oh, D.-W.; Kim, S.; Rogers, J. A.; Cahill, D. G.; Sinha, S. Interfacial Thermal Conductance of Transfer-Printed Metal Films. *Adv. Mater.* **2011**, *23*, 5028−5033.
(40) Cheng, Z.; Yates, L.; Shi, J.; Tadjer, M. J.; Hobart, K. D.; Graham, S. Thermal conductance across β-Ga_2O_3-diamond van der Waals heterogeneous interfaces. *APL Mater.* **2019**, *7*, 031118.
(41) Costescu, R. M.; Wall, M. A.; Cahill, D. G. Thermal conductance of epitaxial interfaces. *Phys. Rev. B* **2003**, *67*, 054302.
(42) Yang, J.; Yang, Y.; Waltermire, S. W.; Wu, X.; Zhang, H.; Gutu, T.; Jiang, Y.; Chen, Y.; Zinn, A. A.; Prasher, R.; Xu, T. T.; Li, D. Enhanced and switchable nanoscale thermal conduction due to van der Waals interfaces. *Nat. Nanotechnol.* **2012**, *7*, 91−95.
(43) Giraud, S.; Kundu, A.; Egger, R. Electron-phonon scattering in topological insulator thin films. *Phys. Rev. B* **2012**, *85*, 035441.
(44) Prasher, R. Acoustic mismatch model for thermal contact resistance of van der Waals contacts. *Appl. Phys. Lett.* **2009**, *94*, 041905.
(45) Chambers, R. G.; Bragg, W. L. The conductivity of thin wires in a magnetic field. *Proc. R. Soc. London A* **1950**, *202*, 378−394.
(46) Xiong, Y.; Lai, N.-C.; Lu, Y.-C.; Xu, D. Tuning thermal conductivity of bismuth selenide nanoribbons by reversible copper intercalation. *Int. J. Heat Mass Transfer* **2020**, *159*, 120077.
(47) Bartkowiak, M.; Mahan, G. D. Heat and electricity transport through interfaces. *Semiconduct. Semimetals* **2001**, *70*, 245−271.
(48) Jin, Q.; Jiang, S.; Zhao, Y.; Wang, D.; Qiu, J.; Tang, D.-M.; Tan, J.; Sun, D.-M.; Hou, P.-X.; Chen, X.-Q.; Tai, K.; Gao, N.; Liu, C.; Cheng, H.-M.; Jiang, X. Flexible layer-structured Bi_2Te_3 thermoelectric on a carbon nanotube scaffold. *Nat. Mater.* **2019**, *18*, 62−68.
(49) Tian, B. Z.; Chen, J.; Jiang, X. P.; Tang, J.; Zhou, D. L.; Sun, Q.; Yang, L.; Chen, Z. G. Enhanced Thermoelectric Performance of SnTe-Based Materials *via* Interface Engineering. *ACS Appl. Mater. Interfaces* **2021**, *13*, 50057−50064.
(50) Zhu, Y.; Han, Z.; Han, B.; Jiang, F.; Wu, X.; Han, C. G.; Deng, Y.; Liu, W. Enhanced Thermoelectric Performance by Strong Phonon Scattering at the Heterogeneous Interfaces of the Mg_2Sn/Mg_3Sb_2 High-Content Nanocomposite. *ACS Appl. Mater. Interfaces* **2021**, *13*, 56164−56170.
(51) Hu, Q.; Qiu, W.; Chen, L.; Chen, J.; Yang, L.; Tang, J. Realize High Thermoelectric Properties in n-Type $Bi_2Te_{2.7}Se_{0.3}/Y_2O_3$ Nanocomposites by Constructing Heterostructures. *ACS Appl. Mater. Interfaces* **2021**, *13*, 38526−38533.
(52) Shi, L.; Li, D. Y.; Yu, C. H.; Jang, W. Y.; Kim, D. Y.; Yao, Z.; Kim, P.; Majumdar, A. Measuring thermal and thermoelectric properties of one-dimensional nanostructures using a microfabricated device. *J. Heat Tranfer* **2003**, *125*, 881−888.
(53) Wingert, M. C.; Chen, Z. C. Y.; Kwon, S.; Xiang, J.; Chen, R. K. Ultra-sensitive thermal conductance measurement of one-dimensional nanostructures enhanced by differential bridge. *Rev. Sci. Instrum.* **2012**, *83*, 024901.
(54) van Setten, M. J.; Giantomassi, M.; Bousquet, E.; Verstraete, M. J.; Hamann, D. R.; Gonze, X.; Rignanese, G. M. The PseudoDojo: Training and grading a 85 element optimized norm-conserving pseudopotential table. *Comput. Phys. Commun.* **2018**, *226*, 39−54.
(55) Paulatto, L.; Fournier, D.; Marangolo, M.; Eddrief, M.; Atkinson, P.; Calandra, M. Thermal conductivity of Bi_2Se_3 from bulk to thin films: Theory and experiment. *Phys. Rev. B* **2020**, *101*, 205419.
(56) BROYDEN, C. G. The Convergence of a Class of Double-rank Minimization Algorithms 1. General Considerations. *IMA J. Appl. Math.* **1970**, *6*, 76−90.
(57) Goldfarb, D. A Family of Variable-Metric Methods Derived by Variational Means. *Math. Comput.* **1970**, *24*, 23−26.
(58) Shanno, D. F. Conditioning of Quasi-Newton Methods for Function Minimization. *Math. Comput.* **1970**, *24*, 647−656.

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